



PID-CONTROLLED SELF-BALANCING ROBOT

EE491 / EE492 Graduation Project

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ABSTRACT

This project presents the design and implementation of a cascade PID controlled self balancing robot on an ESP32 microcontroller and MPU6050 six axis IMU. A dual core FreeRTOS architecture physically isolates the 250 Hz balance task from the 20 Hz wireless telemetry task (Core 0, priority 1), guaranteeing deterministic control-loop execution free from wireless stack preemption. Angle estimation uses a complementary filter gyroscope and accelerometer data. The cascade PID separates angular stabilization from velocity control.

INTRODUCTION & MOTIVATION

Self balancing robots exemplify the classic inverted pendulum an underactuated system with three degrees of freedom (longitudinal displacement, heading, body tilt) but only two actuators. The tilt axis is passive: without active control, any deviation from equilibrium leads to a guaranteed fall due to gravitational restoring torque.

This platform underpins personal micro mobility devices, autonomous guided vehicles (AGVs) in warehouse and manufacturing environments, and urban service robots. The project directly aligns with SDG 9 (Industry, Innovation and Infrastructure) and SDG 11 (Sustainable Cities and Communities) by contributing a low-cost, open-source reference design for efficient transportation.

Goal: A fully validated real time cascade PID balance system with wireless PID tuning and driving capability.

PROBLEM DEFINITION

System dynamics impose four strict real-time constraints:

Control rate >= 100 Hz: Gravitational instability requires fast correction. This design runs at 250 Hz (4 ms period) via vTaskDelayUntil().
Motor pulse frequency: NEMA17 in 1/8 microstepping requires STEP pulses at up to 25 kHz per motor
IMU at 250 Hz: I2C at 400 kHz must complete a full 6 axis read within each 4 ms control window.
Fault protection: Tilt exceeding +/-40 degrees for 25 consecutive cycles triggers FAULT state, cutting motor power.

HARDWARE DESIGN

ESP32 Microcontroller Dual-core 240 MHz Xtensa LX6 520 KB SRAM 4 hardware timers Wi-Fi 802.11b/g/n BT 4.2 Core 1: Balance (250 Hz) Core 0: Telemetry	MPU6050 IMU (GY-521) 6-axis MEMS (triaxial accel + gyro) +/-2g accelerometer DLPF @ 42 Hz +/-250 deg/s gyro 131 LSB/(deg/s) I2C fast-mode 400 kHz
2x NEMA17 + DRV8825 1.8 deg/step 200 steps/rev 1/8 microstepping = 0.225 deg/step Vref = 0.55 V ==> I_max = 1.1 A Heatsinks Step freq. up to 25 kHz	Power & Chassis 3S LiPo 11.1 V / 2200 mAh 100 uF capACITOR 3D-printed chassis High center of mass Total BOM: ~1,900 TL

SOFTWARE ARCHITECTURE

FreeRTOS manages two core-pinned tasks:

BalanceTask (Core 1, 250 Hz): Reads MPU6050 via I2C, applies 5-tap moving average, runs complementary filter, executes cascade PID, updates motor speeds. vTaskDelayUntil() guarantees 4 ms period independent of execution time.
TelemetryTask (Core 0, 20 Hz): Non blockin UDP packet receives PID commands and drive commands.
50 kHz HW Timer ISR (IRAM_ATTR): Generates STEP pulses via accumulator pattern — counter += |speed|; toggle if counter >= 2000 — achieving sub-microsecond jitter at 25 kHz/motor, fully decoupled from FreeRTOS scheduler.
Shared state (speed_L, speed_R, currentSpeed) protected by portENTER_CRITICAL / portEXIT_CRITICAL.

SENSOR FUSION & CALIBRATION

$robotAngle = 0.99 \times (robotAngle + gyroRate \times dt) + 0.01 \times accAngle$

Tilt: $\theta = \text{atan2}(a_z, a_x) \times 57.296 \text{ deg}$ (gravity on X-axis upright; pitch on Z-axis).

Gyro weight 0.99: fast transient response, drift-prone long-term.

Accel weight 0.01: slow but drift-free absolute reference. Time constant τ approx 0.4 s at $dt = 4 \text{ ms}$.

7-Step Boot Calibration: (1) 1.5 s oscillator settle, (2) zero offset registers, (3) 50 warm-up reads to fill DLPF pipeline, (4) wait for $|\theta| < 20 \text{ deg}$ upright pose, (5) collect 300 samples with vibration rejection, (6) write hardware registers [Acc /8, Gyro /4], (7) compute angleOffset from 20 post-cal reads.

Runtime gyro bias tracking via exponential moving average compensates thermal drift during stationary holds ($|\text{error}| < 1.2 \text{ deg}$, $\text{speed} < 80$).

CASCADE PID CONTROL ALGORITHM

A two-level cascade architecture decouples angular stabilization from velocity control, eliminating the coupling that affects single-loop designs. The inner loop executes 12.5x faster than the outer loop.

INNER LOOP — Angle PD (250 Hz)

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angle_error = target_angle - (robotAngle - angleOffset)
acceleration = Kp_st x angle_error - Kd_st x gyroRate
currentSpeed += acceleration
```

OUTER LOOP — Speed PI (20 Hz)

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speed_error = throttle x 3.0 - estimated_speed
speed_integral += speed_error x dt [clamped +/-1000]
target_angle = Kp_thr x speed_error + Ki_thr x integral [clamped to +/-4.5 deg]

Kp_thr = 0.016 (friction against positional drift)
Ki_thr = 0.1 (eliminates steady-state drift over time)
Integral bled at 0.85x per cycle when throttle = 0
```

Parameter	Value	Loop	Physical Effect
Kp_st	3.5	Inner PD · 250 Hz	Angular spring stiffness
Kd_st	0.13	Inner PD · 250 Hz	Gyro-rate damping
Kp_thr	0.016	Outer PI · 20 Hz	Velocity proportional gain
Ki_thr	0.1	Outer PI · 20 Hz	Integral drift elimination

STATE MACHINE & WIRELESS INTERFACE

INIT Calibration running Motors DISABLED	BALANCING PID active · 250 Hz Motors ENABLED	FAULT Tilt > 40 deg Motors DISABLED
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Transitions: INIT -> BALANCING when $|\text{angle error}| < 2 \text{ deg}$ | BALANCING -> FAULT when $|\text{angle}| > 40 \text{ deg}$ for 25 cycles FAULT -> INIT on UDP reset 'R'.

Wi-Fi softAP (SSID: "Denge_Robotu_WIFI") on UDP port 8080. Commands: P/D/T/I (PID gains), Y (throttle), X (steering), O (angle offset), R (reset). Bluetooth permanently removed — shared 2.4 GHz antenna caused 400 ms UDP blackouts under coexistence arbitration.

SYSTEM PERFORMANCE

250 Hz Control Loop Rate	±15 deg Disturbance Recovery	20 Hz Telemetry Rate
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BalanceTask operating at 250. Motor STEP pulses confirmed at 25 kHz/motor at maximum speed with sub microsecond jitter. Complementary filter maintains balance setpoint within +/-0.5 deg over extended runs. Both DRV8825 ICs stay in safe thermal limits at Vref = 0.55 V with heatsinks. UDP telemetry at 20 Hz shows zero packet loss after Bluetooth removal.

CONCLUSION

This project delivers a fully validated cascade PID self-balancing robot with a dual-core real-time architecture. Key contributions: (1) 7-step decoupled IMU calibration eliminating boot-time offset and thermal drift without user intervention; (2) 50 kHz accumulator-based stepper pulse generation with sub-microsecond jitter; (3) physically isolated FreeRTOS dual-core design enabling simultaneous 250 Hz control and 20 Hz telemetry; (4) three embedded debugging case studies reusable for future motor control projects.

FUTURE WORK

(1) Closed-loop position control: Add quadrature encoders to NEMA17 shafts for accurate odometry and position-hold mode.
(2) ROS 2 / micro-ROS integration: Modular firmware is compatible with micro-ROS, enabling multi-robot autonomous navigation.
(3) LQR control: Replace cascade PID with a Linear Quadratic Regulator from the linearized state-space model for improved disturbance robustness.
(4) Adaptive microstepping: Dynamic 1/8 <-> 1/32 switching reduces idle power while preserving low-speed smoothness.
(5) Terrain adaptation: Downward distance sensor for slope detection and pre-compensation.